Aishwarya Teke

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Professional Summary

Associate Data Scientist with 3+ years of experience in building data extraction pipelines, document processing systems, and NLP-based solutions. Skilled in Python, OCR, and text processing techniques for extracting structured insights from unstructured sources such as emails and PDFs. Experienced in developing scalable solutions and transitioning towards Generative AI and LangGraph-based systems.

Skills

Programming: Python, SQL
Data Processing: Pandas, NumPy, Regex
NLP & OCR: Text Extraction, Tesseract OCR, Basic NLP
Tools: VS Code, Jupyter Notebook, Streamlit
Familiar With: LangChain, LangGraph, LLMs, RAG

Work Experience:

PROJECT 1: EMAIL & PDF DATA EXTRACTION SYSTEM

- Developed automated pipelines to extract structured data from emails and PDF documents
- Implemented text parsing and regex-based extraction for key fields
- Processed scanned PDFs using OCR techniques
- Built preprocessing workflows for cleaning and structuring raw data
- Reduced manual effort and improved data accuracy.

PROJECT 2: MEDICAL CHATBOT ASSISTANT-USING Generative AI (RAG):

Problem Statement: Health professionals and patients needed quick, reliable access to medical insights from unstructured data and research papers. Manual searches were slow and error-prone, prompting the creation of a trusted, citation-backed medical chatbot.

Solution: Built a **RAG** pipeline **with LangChain and GPT-4 Turbo**, integrated with Pinecone for retrieving relevant medical literature.

- Fine-tuned GPT-4 Turbo using **LoRA** adapters for medical Q&A, minimizing hallucinations through reranking and factual grounding.

- Integrated **Streamlit** to build an intuitive, user-friendly interface for smooth **chatbot** interaction.
- Utilized GPT-4 Turbo to generate empathetic, context-rich responses by blending retrieved knowledge with model insights.
- Stored conversation history in **Pinecone vector DB** to preserve context and enhance retrieval for continuous dialogue.
- Achieved **84% accuracy** in context-aware, reference-backed medical responses, **reducing manual research time by 35%.**

PROJECT 3: RESUME SCREENING ASSISTANCE USING RETREIVAL AUGMENTED GENERATION (RAG)

Problem Statement: Manual resume screening is slow, inconsistent, and error-prone, making it hard for HR to match candidates with job requirements efficiently. Large volumes of resumes make the process time-consuming and difficult to scale while ensuring unbiased shortlisting.

Solution: Built a Retrieval-Augmented Generation (RAG) pipeline using **LangChain + GPT-4 Turbo**, integrated with Pinecone vector DB to fetch relevant match. **OCR (Optical Character Recognition)** used to extract text from scanned resumes if any.

- Store resume embeddings in a **pinecone vector DB**.
- LLM **GPT-4** retrieved top-matching resumes based on job description query and summarized candidate fit, skills match and gives ranking/score.
- Built a user-friendly interface using **Streamlit** for uploading resumes and querying job-fit analysis.
- Designed modular **FastAPI** services for retrieval, inference, and scoring pipelines.
- Measured performance using precision, recall on retrieved chunks and user validation of generated summaries.

Outcome: The RAG-powered resume screening solution **reduces manual screening time by up to 80%** and **improves candidate-job matching accuracy to around 90%**, accelerating the hiring process. It ensures a **scalable, consistent, and fair** approach to shortlisting, enhancing overall HR efficiency.

Tools used: Python, Hugging Face, LangChain, Pinecone, GPT-4.

PROJECT 4: RISK MANAGEMENT SYSTEM:

Problem Statement: Fraudulent vehicle insurance claims cause significant financial losses and are hard to detect manually. Machine

learning system is needed to accurately identify suspicious claims in real-time, reducing risk and improving claim processing efficiency**.**

Solution: Built a complete machine learning solution to identify fraudulent vehicle insurance claims, minimizing financial losses for the insurer.

- Conducted thorough data analysis on large datasets, including data cleaning, **KNN-based imputation** for missing values, and outlier detection using **Z-score and Boxplots.**
- Performed feature engineering and encoded categorical variables using **One-Hot and ordinal** Encoding;
- Addressed class imbalance with **SMOTE (Synthetic Minority Over-sampling Technique)**.
- Developed and assessed multiple models **(Logistic Regression, Decision Tree, Random Forest, Boost)** choosing XGBoost for its **87% accuracy**.
- Optimized model performance via hyperparameter tuning with **GridSearchCV** and evaluated results using **Precision, Recall, and F1-score**.
- Deployed the trained model through a **Flask API** and validated it with **Postman** for real-time fraud detection.

Education

M.Tech in Computer Engineering – DBATU
B.E. in Computer Science – Shivaji University